AMERY ICE SHELF BOREHOLE MICROCAT AND CTD DATA, 2009/2010 SEASON - DATA PROCESSING AND QUALITY

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INTRODUCTION

This report summarises the latest AMISOR oceanographic data, season 2009/2010, collected from oceanographic equipment deployed through the Amery Ice Shelf boreholes. Time series data of temperature, salinity and pressure were downloaded from two strings of Seabird SBE37IM microcats at AMISOR boreholes AM03 (12 months of data) and AM04 (13 months of data days of data). CTD casts, using an FSI 3" MicroCTD, were taken through new boreholes at AM05 and AM06. Data processing and data quality details are included in the report.

Table 1: Station details for AMISOR season 2009/2010 borehole microcats. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast. Instrument pressures/depths are mean values over the data dates. Note that data were not downloaded from microcats 1973 and 1974 at AM04, due to lack of time at the site.

Borehole microcats

borehole	position at given date		microcat	instrument pos.		data dates		rec.	clock
	latitude	longitude		depth (m)	pressure (dbar)	start	end	interval (min.)	error (sec.)
AM03	70°32.851'S	70°20.382'E	4055	652.3	660.0	28-Jan-2009	29-Jan-2010	30	+193
AM03	(on 29-Jan-2010)		4054	860.1	870.7	28-Jan-2009	29-Jan-2010	30	+155
AM03	"	"	3883	1209.4	1225.4	28-Jan-2009	29-Jan-2010	30	+108
AM04	69°52.962'S	70°19.478'E	1972	528.7	534.8	30-Jan-2009	04-Mar-2010	30	+201
	(on 04-Mar-2010)			539.2	545.4	(before 09/07/2009 i.e. before start of P step)			
				521.6	527.6	(after 11/07/2009 i.e. after end of P step)			
AM04	"	"	1973	-	-	-	-	30	+242
AM04	"	"	1974	-	-	-	-	30	+134

MICROCAT DATA

The microcats were downloaded by Adam Christensen, and a complete description of the relevant field operations is given in the report:

AMISOR Operations - 2009/2010, Adam Christensen, Australian Antarctic Division unpublished report, 11pp.

The microcat moorings have already been described in previous field and data reports. A year's worth of data was successfully downloaded from the 3 microcats at AM03 (originally deployed in December 2005). For AM04 (deployed in January 2006), time ran short at the site and only 1 of the 3 microcats (serial no. 1972) was downloaded (13 months of new data). Data from the other 2 microcats at AM04 (serial nos. 1973 and 1974) will hopefully be downloaded next season. An attempt was made to download data from the microcats at borehole AM01, but this failed due to flat microcat batteries (an expected outcome from the previous season's download). No attempt was made to communicate with the microcats at borehole AM02, as the microcat battery life had ended two seasons previous. All latest microcat data downloaded during the 2009/2010 season (Table 1) were assigned the same decimal time scheme consistent with previous data i.e. decimal days as counted from midnight on

December 31st 2000 (so, e.g. midday on January 1st 2001 = 0.5 decimal time; midday on January 1st 2002 = 365.5).

The microcats were downloaded using the SeaBird terminal program Seaterm, and instrument clock errors were noted at the time (Table 1). Clocks were reset to GPS time for all 4 downloaded instruments, following data download and prior to reinitialisation of logging. For the 2 microcats which were not downloaded (serial nos. 1973 and 1974), clock error was checked, however it is unknown whether the clocks were reset to GPS time - the answer to this should be obvious from the magnitude of the clock error at the next download.

Manufacturer supplied calibrations (June 2005 for AM03 instruments [except October 2004 for microcat-3883 pressure], February 2003 for AM04 instruments) were applied internally by the microcats, and calibrated data were output. The data were then processed, as follows:

* The files were padded at the start and end, data gaps were checked for and filled, and decimal times were calculated.

* Decimal times were "compressed" linearly throughout the time series to correct for clock error (Table 1).

* Mean pressure/depth values were calculated for the year's data for each microcat (Table 1).

Microcat data quality notes

The following bad data were removed from the files:

* AM03, all 3 microcats - the first 15 data records occurred at odd times, and must have been instrument tests. Good data commences from 0130 on 28/01/2009.

* AM03, microcat 4054 - conductivity/salinity data spike removed for data point 8220

* AM04, microcat 1972 - conductivity/salinity spikes removed for data points 2994-2995, and 5957

For microcat 4055 at AM03, several data/comm. errors occurred, resulting in data gaps for the following times: 1330 on 08/04/2009 to 0100 on 21/04/2009
2130 on 11/05/2009 to 0900 on 24/05/2009
0530 on 11/11/2009 to 1700 on 23/11/2009

Microcat data are plotted in Figures 1 to 3. The large salinity spikes evident in Figure 1 for microcat 4055 (AM03) are most likely due to fouling (ice crystals or critters). For AM03, the pressure variation throughout the time series is very similar for all 3 microcats (Figure 3), with all pressures getting shallower by ~0.5 dbar over the second half of the times series. Although the order of this drift is of the same order as the expected pressure sensor drift as quoted by the manufacturer, the strong correlation in drift between the 3 instruments suggests that the drift is real.

For AM03, numerous strong current events occur (particularly in the second half of the times series), causing temporary "push-up" of the instrument string (Figure 3). For the shallowest microcat (serial 4055), there are 2 push-up events evident around decimal days ~3212 and ~3224 (Figure 3). The first of these events is visible in the other 2 microcats, however the second is not - if this event is real (unknown), then it must be occurring over such a small time scale that it is missed by the other 2 instruments, due to time mismatch from the clock drift errors.

For the single time series of microcat 1972 at AM04, a push-up event of ~4 dbar occurs from decimal day ~3073 to ~3076, while a large step of ~15 dbar commences decimal around day ~3111, taking ~3 days to reach the 15 dbar magnitude (Figure 3). Pressure remains at this shallower level for the remainder of the record. Confirmation of these pressure changes await correlation with data from the other 2 microcats, hopefully to be downloaded next season. However, note that a similar temporary pressure step of ~15 dbar occurred the previous year, specifically: for microcat 1972, pressure gets

shallower by 15 dbar from 27/06/2008 to 07/07/2008; for the latest data for microcat 1972, pressure gets gradually shallower by 15 dbar from 09/07/2009 to 11/07/2009, then remains at this pressure. In addition, for the latest data a temperature step is evident at the same time as the pressure step (Figure 2). This evidence suggests the step is real and caused by local deformation of the iceshelf, effectively lifting the instrument string.

For AM04, pressure steadily drifts shallower by ~1 dbar over the time series prior to the 15 dbar step; after the step, the rate of drift increases to ~3 dbar over the remaining 8 months of data. Again, correlation with the pressures from the other 2 microcats is required, however it is likely that all these pressure drift features are real.

From conductivity and temperature sensor calibration drift estimates in a previous data report for 2003/2004 data (i.e. 0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature), we might expect the following calibration drifts at the time of the latest 2009/2010 data downloads:

AM03 (in water for 4 years) total S drift = 0.014 (PSS78) total T drift = 0.0020^{o}C
AM04 (7 years since calibration) total S drift = 0.025 (PSS78) total T drift = 0.0035^{o}C
(noting that AM04 conductivity and temperature sensors were calibrated in February 2003).

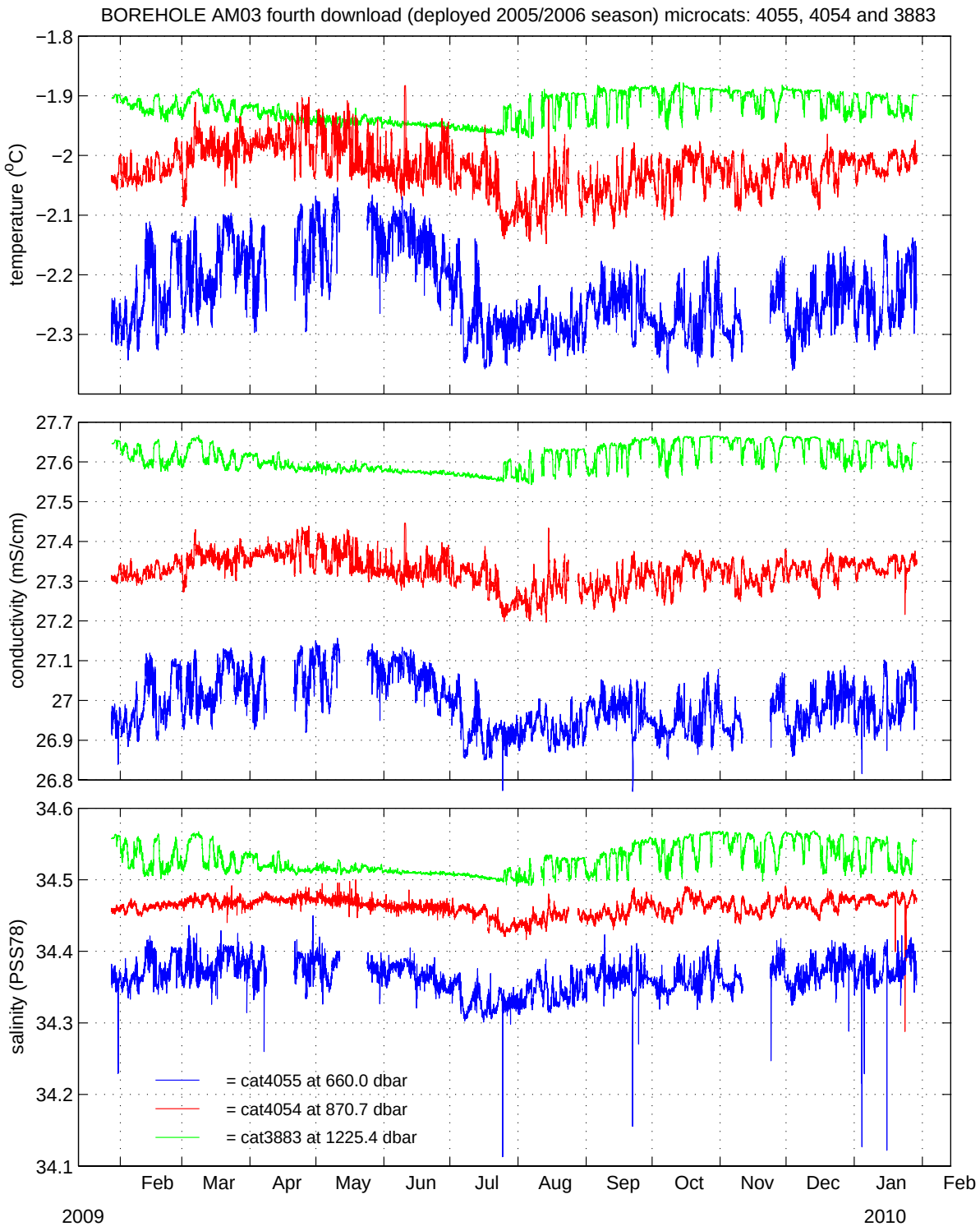


Figure 1: Borehole AM03 microcat data downloaded in 2009/2010 season.

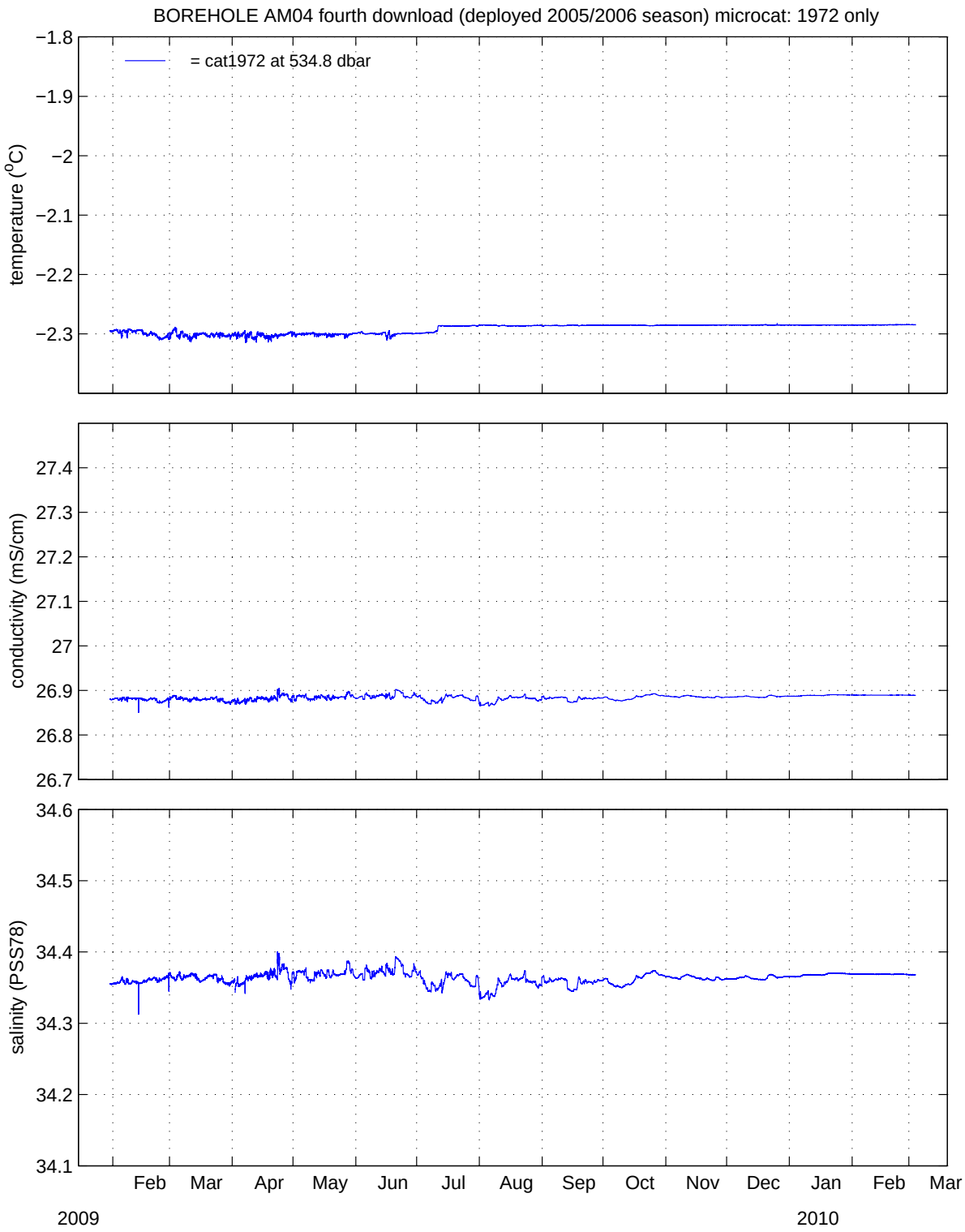


Figure 2: Borehole AM04 microcat data downloaded in 2009/2010 season.

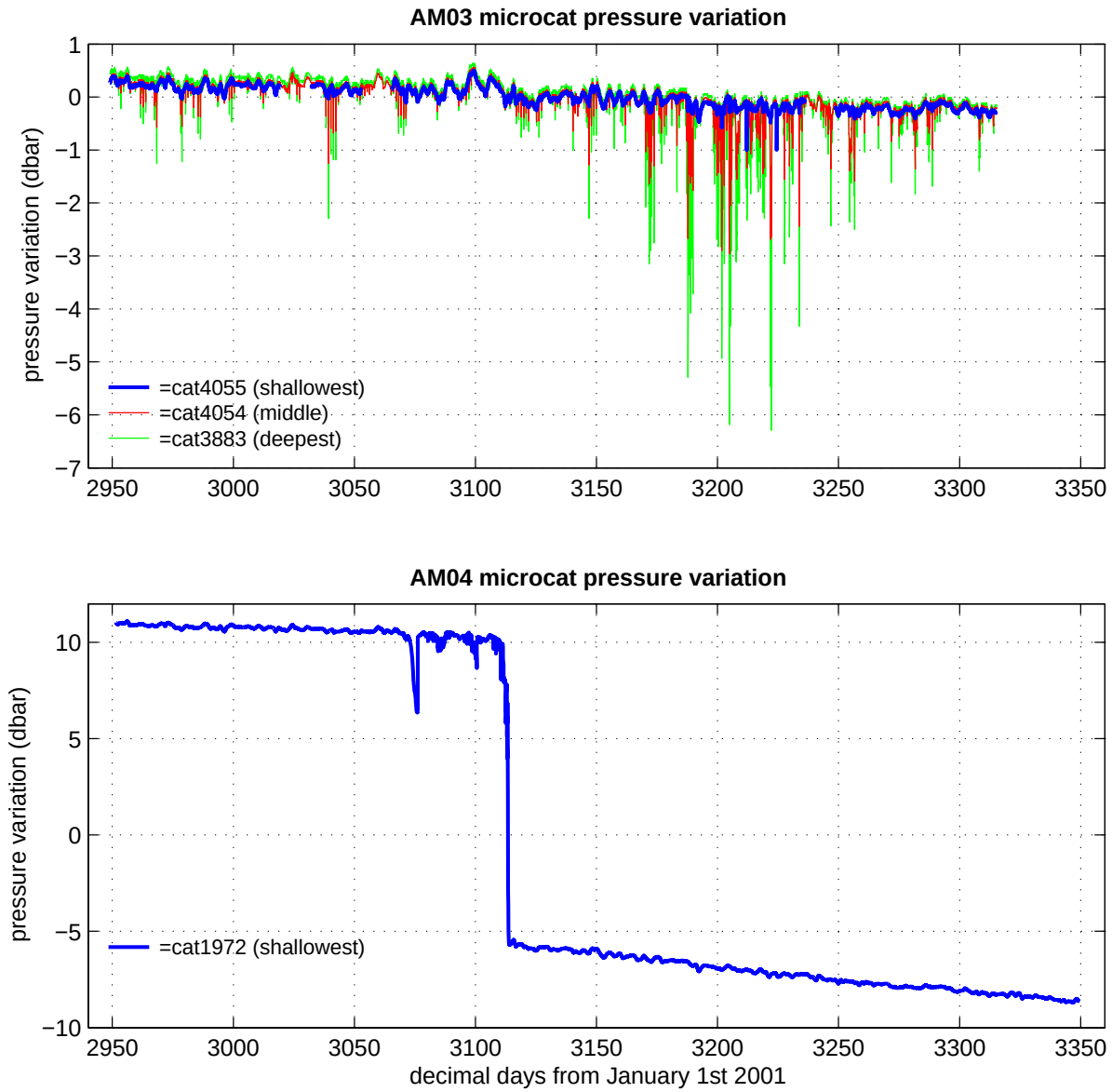


Figure 3: Pressure sensor data from boreholes AM03 and AM04 microcat data downloaded in 2009/2010 season. Plotted pressures are variations about the mean value.

CTD DATA

FSI CTD raw data and field notes were provided by Al Elcheikh. Pre-season laboratory calibrations of the FSI CTD temperature, pressure and conductivity sensors were done at CSIRO Division of Marine Research (19th August 2009 for temperature and conductivity, 26th August 2009 for pressure). In the field, data were logged direct to PC via a seacable, and "raw" data were output from the CTD in engineering units, with CSIRO calibration coefficients applied for temperature, pressure and conductivity. The bottom contact switch was fitted, but mostly did not function. At borehole AM05, the data were output from the instrument at 1.83 Hz, except for cast number 5 which was output at 2.58 Hz; at borehole AM06, data were output at 2.58 Hz. Note that no in situ calibration checks were available for the season's CTD work - no Niskin bottles were deployed through the iceshelf, and no shipboard calibration check was done for the CTD (no suitable cruises scheduled). Therefore the data accuracy relies soley on the pre-season CSIRO lab. calibration.

Communication problems resulted in numerous corrupted data scans, however most of the data were retrievable - manual editing was required. A more serious problem occurred with pressure, with pressure discontinuities and false readings occurring for many casts. The errors are in many cases assumed due to blockage of the pressure port by ice crystals, however false pressure data occur for a few cases e.g. for cast 6 at AM05, all pressure data appear to be out by ~100 dbar. Blocks of bad conductivity data are also evident, in particular the first ~715 dbar for cast 4 at AM05. Note that bad conductivity data have been observed in previous seasons (e.g. data from AM02 in 2000/2001).

Table 2: CTD cast details for Amery Ice Shelf Boreholes AM05 and AM06. Note the following: borehole depth=distance from top of borehole to base of iceshelf; depth to water surface=distance from top of borehole down to water surface in the borehole; bottom depth=total water depth from water surface to ocean bottom; max.P=maximum pressure of CTD cast; elevation=CTD elevation above bottom at the bottom of the cast. Note that for AM05, casts 5, 6 and 7 were part of the same CTD deployment; for AM06, casts 1, 2 and 3 were part of a first deployment, while casts 4 and 5 were part of a second deployment.

bore -hole	cast no.	UTC start time	UTC start date	latitude	longitude	borehole depth (m)	depth to water surface (m)	bottom depth (m)	max.P (dbar)	elev. (m)
AM05	1	0607	09-Dec-2009	70°13.9'S	69°40.5'E	624	71.1	908	527	387
AM05	2	2313	09-Dec-2009	70°13.9'S	69°40.5'E	624	70.0	909	494	421
AM05	3	0325	10-Dec-2009	70°13.9'S	69°40.5'E	624	69.7	909	911	9
AM05	4	1402	10-Dec-2009	70°13.9'S	69°40.5'E	624	70.5	909	908	12
AM05	5	0351	12-Dec-2009	70°13.9'S	69°40.5'E	624	70.1	909	904	16
AM05	6	1142	12-Dec-2009	70°13.9'S	69°40.5'E	624	~70	909	?	?
AM05	7	1703	12-Dec-2009	70°13.9'S	69°40.5'E	624	~70	909	875	45
AM05	8	0619	14-Dec-2009	70°13.9'S	69°40.5'E	624	70.2	909	?	?
AM06	1	1222	01-Jan-2010	70°14.165'S	71°21.829'E	607	73.4	829	807	32
AM06	2	1448	01-Jan-2010	70°14.165'S	71°21.829'E	607	~73	829	787	51
AM06	3	1549	01-Jan-2010	70°14.165'S	71°21.829'E	607	~73	829	781	57
AM06	4	0501	02-Jan-2010	70°14.165'S	71°21.829'E	607	74.7	827	802	35
AM06	5	1002	02-Jan-2010	70°14.165'S	71°21.829'E	607	~74	827	819	18

FSI CTD processing and calibration

Two data sets have been produced for the borehole CTD data: 2 dbar averaged downcast and upcast data; and the complete data as logged at ~2 Hz.

The following processing steps were followed to obtain all data:

* Bad data scans were repaired or removed from all files.

* The multiple files for each cast were inspected then either rejected or retained - retained files for each cast were then concatenated to form a single file for the cast (Table 3).

Table 3: Concatenation details for multiple raw files from each CTD cast. "renamed concatenated file" is the renamed raw file for each cast, comprising the retained original raw files; "raw file" is each of the original multiple raw files for each cast, with content as indicated, and either retained or rejected in the renamed concatenated raw file (as indicated).

borehole	*renamed concatenated file*	*original raw file*	*content*	*action*
AM05	am05d001	am05d001	out of water pre deployment	reject
	"	am05d001a	downcast 0-527 dbar	retain
	"	am05_test_to_get_past_constriction	upcast 527-513 dbar	reject
	"	am05d001b	tiny bit at 515 dbar	reject
AM05	am05d002	am05d002	out of water pre deployment	reject
	"	am05d002a	downcast 0-494 dbar	retain
		am05d002_upcast_test	tiny bit at 494 dbar	reject
AM05	am05d003	am05d003	out of water pre deployment	reject
	"	am05d003a	downcast 0-911 dbar	retain
		am05d003a_upcast	tiny bit at 911 and 907 dbar	reject
		am05d003b	lots of bad pressure data	reject
AM05	am05d004	am05d004	downcast 0-908 dbar	retain
AM05	am05d005	am05d005	downcast 0-904 dbar	retain
	"	am05d005_up	tiny upcast bit near bottom	reject
	"	am05d005a_up	tiny upcast bit near bottom	reject
AM05	am06d006	am05d006	yoyos between ~660-860 dbar	retain
AM05	am05d007	am05d007	mostly downcast, 636-875 dbar	retain
	"	am05d007_up	upcast 875-0 dbar	retain
AM05	am05d008	am05d008	out of water pre deployment	reject
	"	am05d008a	out of water pre deployment	reject
	"	am05d008b	downcast 0-870 dbar then upcast to 240 dbar	retain
AM06	am06d001	am05d001	downcast 0-807 dbar	retain
	"	am05d001_up	upcast 807-681 dbar	retain
AM06	am06d002	am05d002	downcast 685-787 dbar	retain
	"	am05d002_up	upcast 785-677 dbar	retain
AM06	am06d003	am05d003	downcast 666-781 dbar	retain
	"	am05d003_up	upcast 781-571 dbar	retain
	"	am06d003_operation_freedom	upcast 571-0 dbar	retain
AM06	am06d004	am05d004	downcast 0-185 dbar	retain
	"	am05d004a	downcast 185-802 dbar	retain
	"	am05d004_up	upcast 802-541 dbar	retain
AM06	am06d005	am05d005	downcast 541-818 dbar	retain
	"	am05d005a	long upcast for current meter, 818-813 dbar	retain
	"	am05d005b	long hang, ranges between 813 and 816 dbar	retain
	"	am05d005_operation_freedom	slow upcast 819-810 dbar	retain
	"	am05d005_up	upcast 809-520 dbar, yoyo back to 550 dbar	retain

* Raw data from the renamed concatenated files were plotted in matlab for initial quality check. Bad data scans deleted from the concatenated files are listed in Table 4.

* Surface pressure offset was found by averaging the 21 pressure points previous to the CTD entering the water (Table 5). This offset was then removed from FSI pressure data, and salinity recalculated with the new pressure values.

* For the ~2 Hz data files, decimal times were calculated and included in the files.

The following further processing steps were followed to obtain 2 dbar averaged data:

* Separate pressure monotonic files for downcast and upcast data were formed from the raw data.

* 2 dbar averages were formed for temperature, pressure and conductivity from the pressure monotonic downcast and upcast files. The minimum attendance required to form each 2 dbar bin was 2 data points. A salinity value for each 2 dbar bin was calculated from the 2 dbar averages.

Finally, both the 2 dbar data and full ~2Hz data were plotted for a last quality control check. General compatibility of data was examined by comparing the different casts, and by comparing the down and upcast data within each cast. Further bad data found at this stage are included in Table 4. For the ~2 Hz data, CTD pressure data were plotted against pressure data from two other sources: simultaneous current meter data collected on some of the CTD casts (note that current meter data are not discussed in this report); and winch line out readings from the winch control panel (readings in metres recorded manually in the field notes, and converted to pressure in dbar). Both these alternate data sources have been corrected for vertical offset from the CTD pressure sensor. The comparisons are shown in Figures 4 and 5. Note that in the Figures, some bad pressure data (found from initial plotting of the data as discussed above) have already been removed. Large amounts of bad pressure data were identified from the comparison, including much of cast 5 and all of cast 6 from borehole AM05, and all data from the ocean cavity below the borehole for casts 1, 2 and 3 from borehole AM06.

Final 2 dbar averaged CTD profiles, with all bad data removed, are plotted in Figures 6 and 7.

Table 4: Bad data scans deleted from CTD raw (i.e. ~2 Hz) *.dat files.

borehole	*cast no.*	*scan nos.*	*~pressure (dbar)*	*comment*
AM05	1	4855-5051	424-437	clogged pressure
AM05	2	6256-7829	496.5-494	tiny bit of upcast, deleted for 2 dbar averaging only
AM05	4	1-17486	0-715	bad conductivity - delete conductivity/salinity only
AM05	4	25111-74522	908	hanging at bottom for long time, T changes, deleted for 2 dbar averaging only
AM05	5	8719-16651	491-646	bad pressure
AM05	5	17360-23109	656-727	bad pressure
AM05	5	32626-38489	849-890	bad pressure
AM05	5	38873-39376	899-903	bad pressure
AM05	5	39510-40016	~904	bad pressure
AM05	6	1-31933	whole cast	bad pressure
AM05	7	1-3778	636-639	strange little initial yoyo
AM05	7	7948-33672	748-bottom-598	clogged pressure
AM05	7	33673-45176	~570	bad pressure
AM05	7	85710-86688	255-172	clogged pressure
AM05	8	4896-62564	542-870-240	bad pressure, then suspect everything
AM06	1	1-2594	0-14	pressure clogged
AM06	1	9681-20435	all below borehole	bad pressure
AM06	2	4836	748	tiny conductivity spike
AM06	2	1-7287	whole cast	bad pressure
AM06	3	1-31100	all below borehole	bad pressure
AM06	5	97891, 97980	800	2 tiny conductivity spikes

Table 5: Surface pressure offsets for each CTD cast. Note: am05d005, 6 and 7 are part of the same deployment; am06d001, 2 and 3 are part of the same deployment; am06d004 and 5 are part of the same deployment.

cast	*surface p offset (dbar)*	*cast*	*surface p offset (dbar)*
am05d001	0.319	am06d001	0.416
am05d002	0.209	am06d002	0.416
am05d003	0.263	am06d003	0.416
am05d004	0.220	am06d004	0.156
am05d005	0.233	am06d005	0.156
am05d006	0.233		
am05d007	0.233		
am05d008	0.312		

FSI CTD data quality

In general, data accuracy for this season's FSI CTD data must be considered of lower accuracy than for previous seasons, as no in situ calibration checks were performed (i.e. no Niskin bottle salinity samples collected from below the iceshelf, and no shipboard comparison casts with ship's CTD). From shipboard calibration tests of the FSI CTD in previous seasons, typical errors/corrections were found for the FSI CTD sensors (chosing the largest error found):

pressure - error up to 3 dbar over 1000 dbar (from 2000/2001 season)
temperature - error up to 0.005^{o}C (from 2000/2001 season)
conductivity - error up to ~0.21 mS/cm (from 2005/2006 season)
salinity - error upto ~0.3 (PSS78) (from 2005/2006 season)

Note that for conductivity and salinity, the correction from previous seasons was always a negative one. Therefore the uncorrected conductivity/salinity values for the latest 2009/2010 season's data are likely to be too high, though without any in situ calibration checks the magnitude of the inaccuracy is unknown. Care should be taken when comparing the 2009/2010 data to data from previous seasons.

In addition to these sensor inaccuracies, significant pressure sensor errors were encountered during the 2009/2010 season, as described earlier. Fortunately, alternate sources of equivalent pressure data (from the current meter and winch lineout) were available for verification of good CTD pressure data. Unfortunately, significant amounts of bad pressure data were identified, meaning a lot of data have been deleted from the files (Table 4). The bad pressures were in most cases due to an apparently blocked pressure sensor, but there were occasions of unexplained erroneous data. Clearly the CTD is on it's last legs, and needs replacement before any further borehole CTD work. Overall, the entire FSI CTD data set from the 2009/2010 season must be treated with caution.

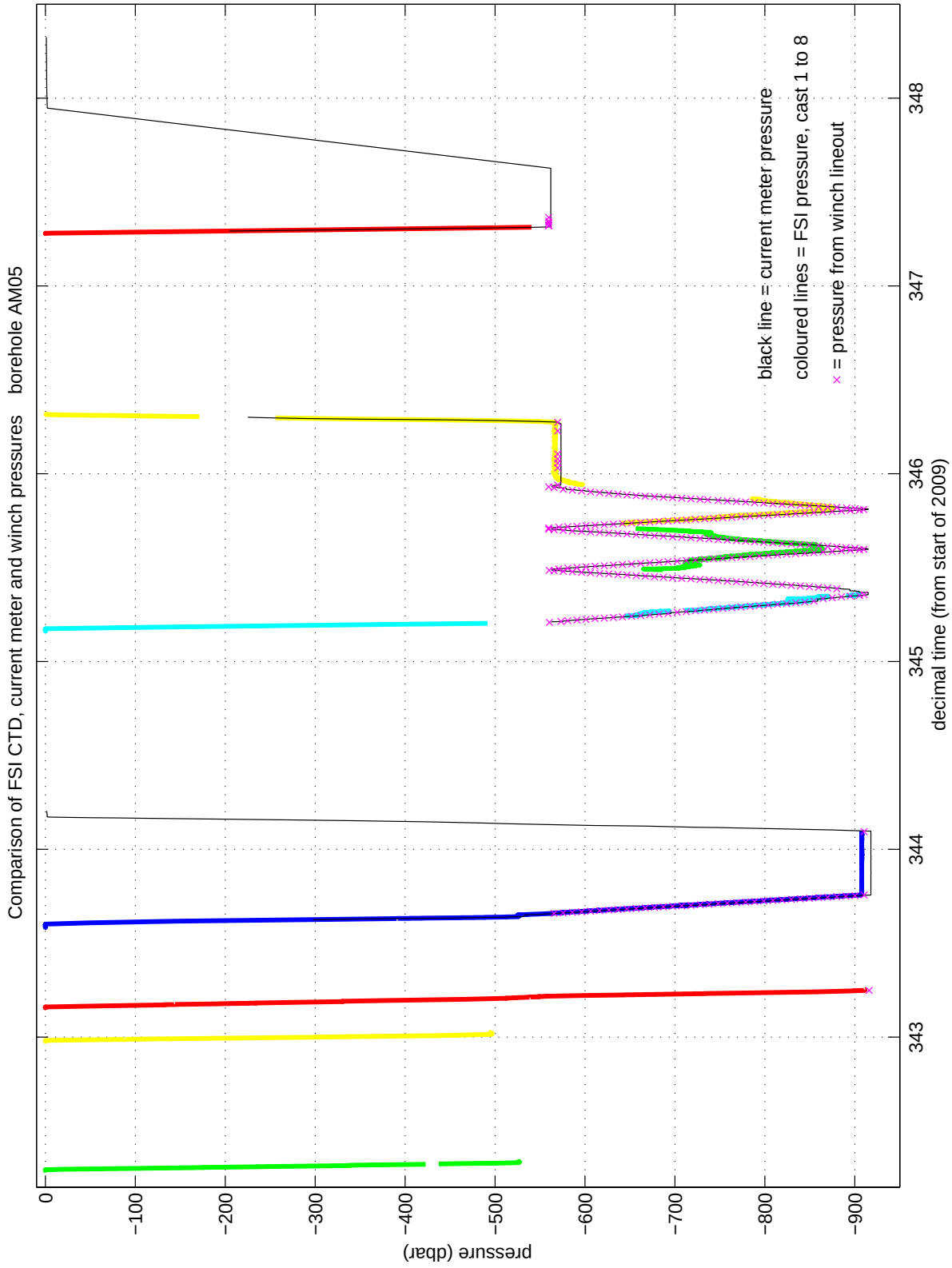


Figure 4: Borehole AM05 comparison of pressure data. Note that winch lineout data have been converted to pressure in dbar from the water surface; and both winch and current meter pressures have been corrected for vertical offset from the CTD pressure sensor. Bottom of the borehole is at ~561 dbar.

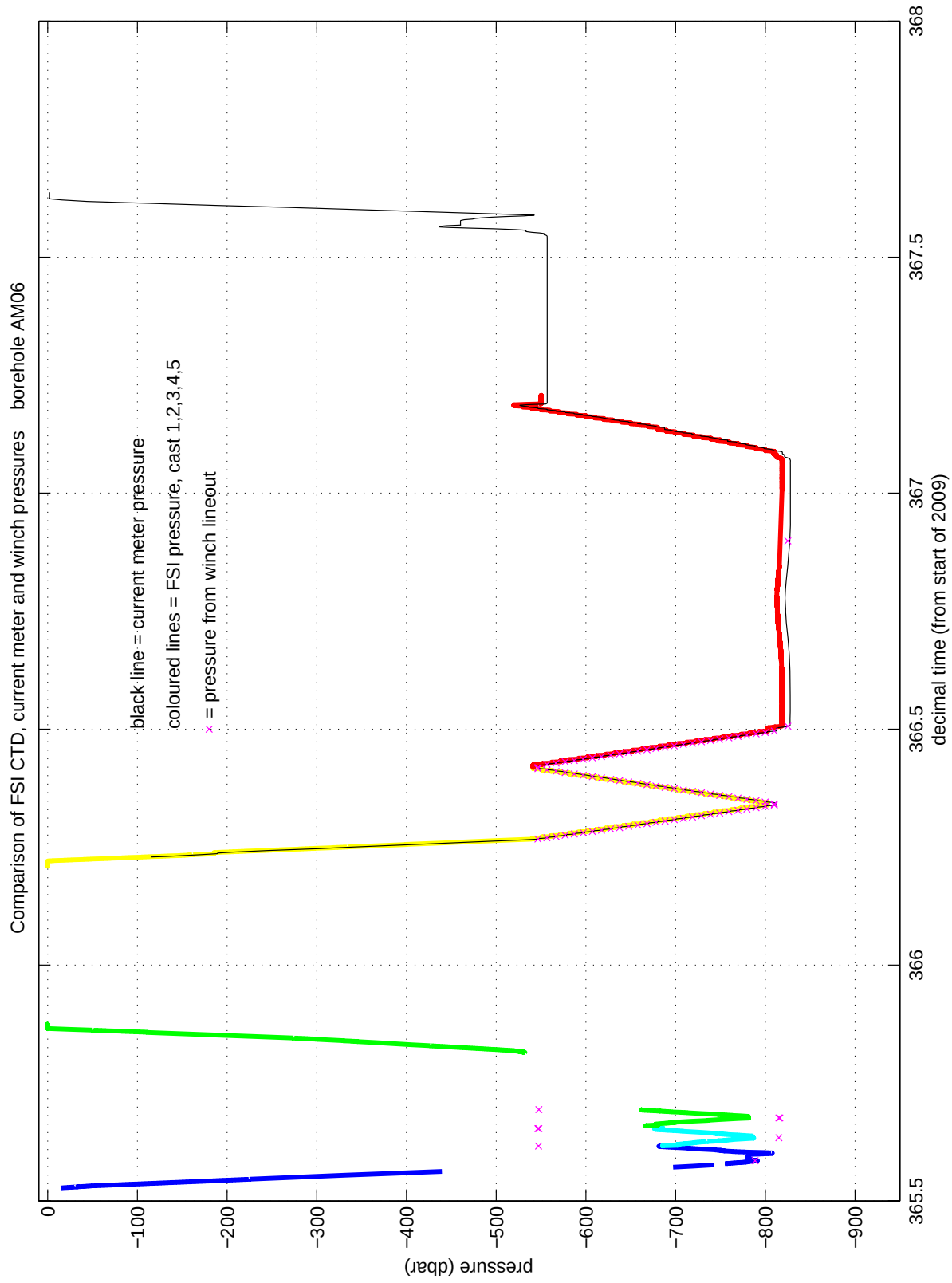
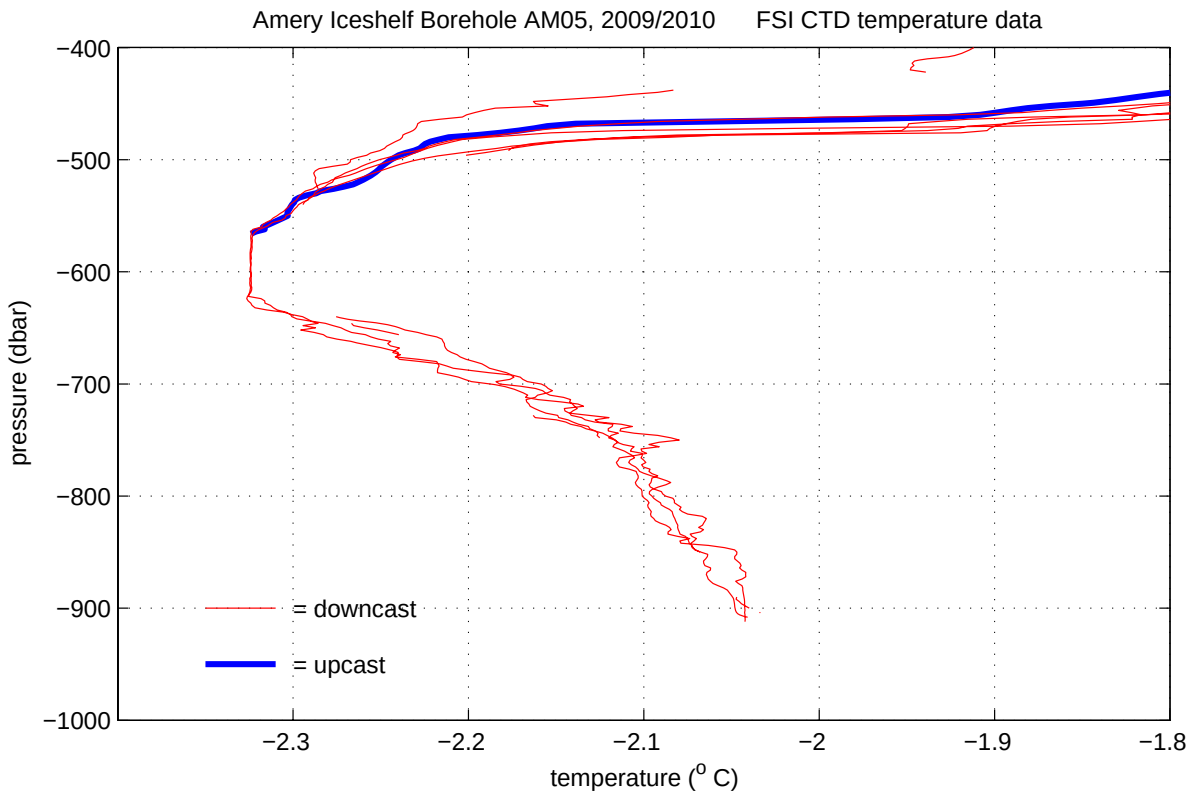


Figure 5: Borehole AM06 comparison of pressure data. Note that winch lineout data have been converted to pressure in dbar from the water surface; and both winch and current meter pressures have been corrected for vertical offset from the CTD pressure sensor. Bottom of the borehole is at ~540 dbar.



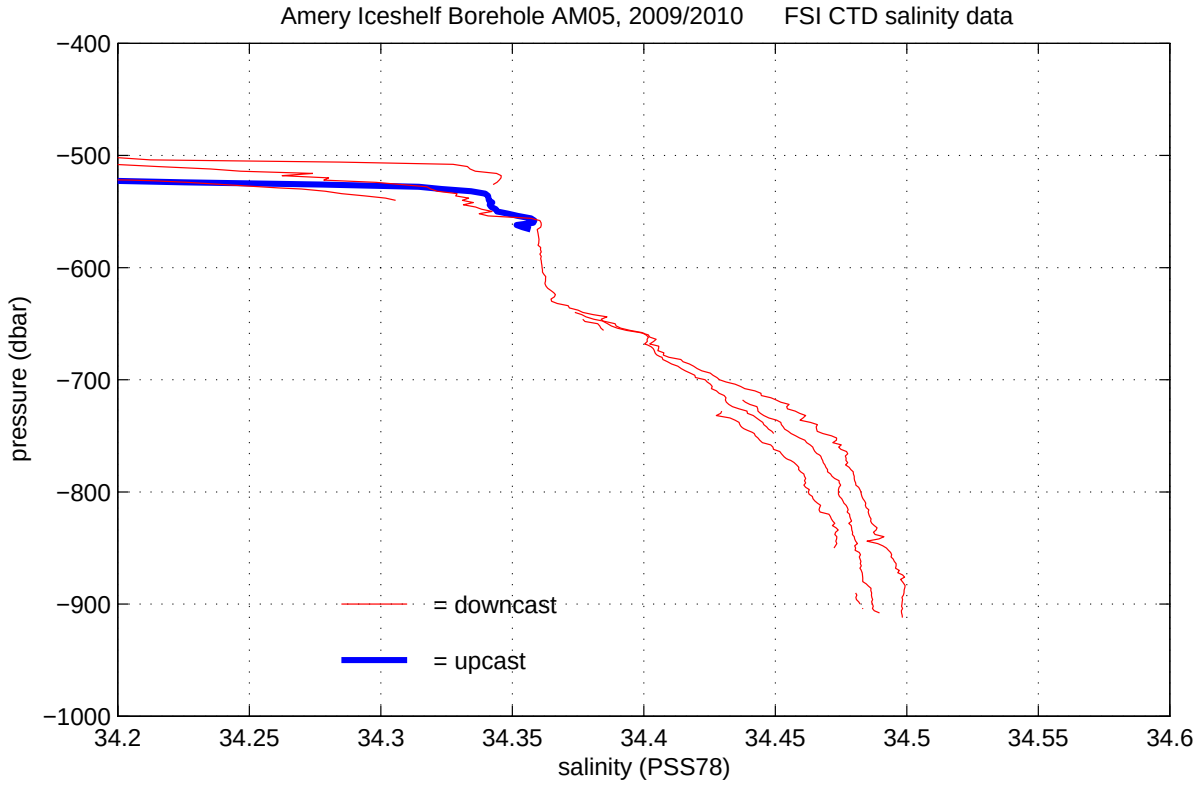
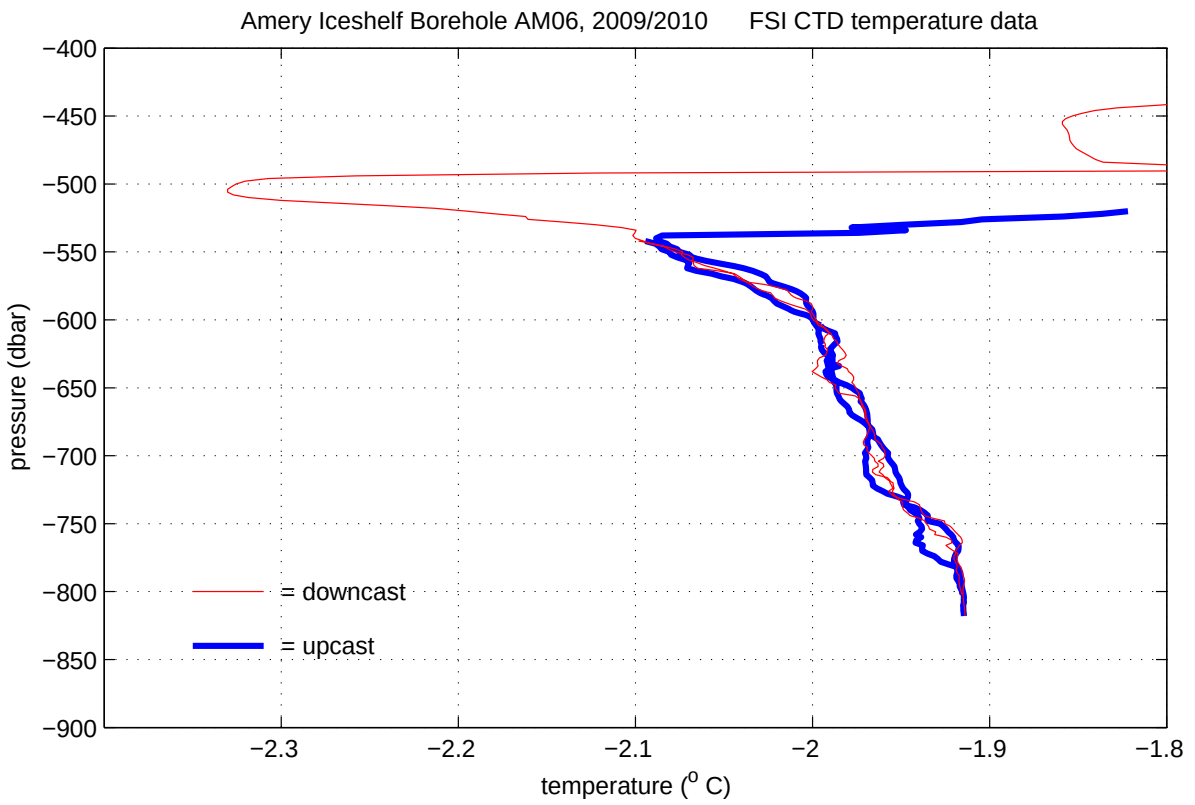


Figure 6: Amery Ice Shelf borehole AM05, CTD 2 dbar averaged data. Bottom of the borehole is at ~561 dbar.



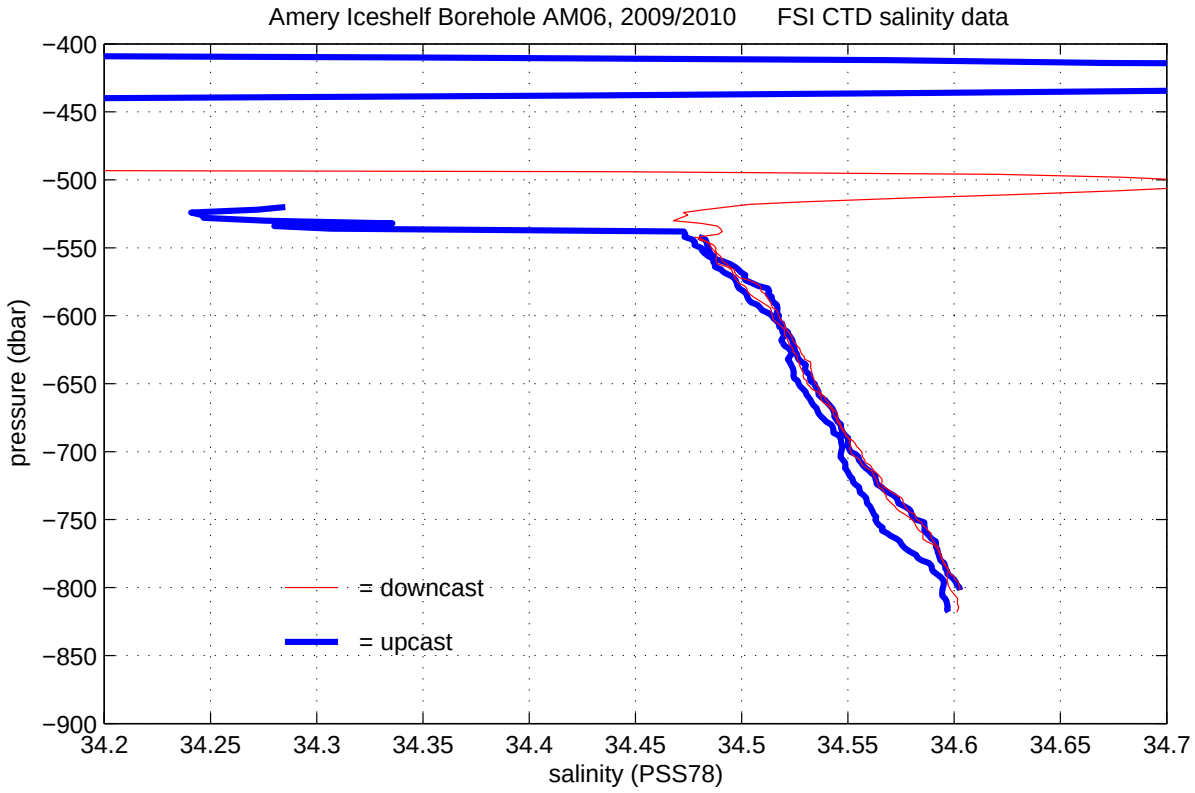


Figure 7: Amery Ice Shelf borehole AM06, CTD 2 dbar averaged data. Bottom of the borehole is at ~540 dbar.

DATA FILES

New microcat data from boreholes AM03 and AM04, and 2 dbar averaged CTD data from boreholes AM05 and AM06, are contained in text and matlab format files, as follows:

1. Microcat data

AM03
Data from the fourth data download (i.e. 28/01/2009-29/01/2010) are contained in the files:
cat4055_4.dat, cat4054_4.dat, cat3883_4.dat (text files), and
cat4055_4.mat, cat4054_4.mat, cat3883_4.mat (matlab files).
READMEcatam03_4 details the microcat file formats.

The updated unified files (i.e. all data from initial deployment time on 17/12/2005, updated with data from the fourth download) are contained in the files:
cat4055_all.dat, cat4054_all.dat, cat3883_all.dat (text files), and
cat4055_all.mat, cat4054_all.mat, cat3883_all.mat (matlab files).
READMEcatam03_all details the microcat file formats.

AM04
Data from the fourth download (i.e. 30/01/2009-04/03/2010) are contained in the files:
cat1972_4.dat (text file), and cat1972_4.mat (matlab file).
READMEcatam04_4 details the microcat file formats.

The updated unified files (i.e. all data from initial deployment time on 11/01/2006, updated with data from the fourth download) are contained in the files:
cat1972_all.dat (text file), and cat1972_all.mat (matlab file).
READMEcatam04_all details the microcat file formats.

2. CTD data

AM05
Data from the CTD casts at borehole AM05 are contained in the files:

fsiam0500x.dwn_av	cast x (x=1 to 8) downcast 2 dbar averaged data, text files
fsiam0500x.upp_av	cast x (x=1 to 8) upcast 2 dbar averaged data, text files
fsiam05.mat	downcast and upcast data for all casts, matlab file
fsiam0500x.ful	cast x (x=1 to 8) complete data at the full ~2Hz data rate, text files
fsiam0500xful.mat	cast x (x=1 to 8) matlab version of the full ~2Hz data
READMEam05_ctd	a README file detailing file formats

AM06
Data from the CTD casts at borehole AM06 are contained in the files:

fsiam0600x.dwn_av	cast x (x=1 to 5) downcast 2 dbar averaged data, text files
fsiam0600x.upp_av	cast x (x=1 to 5) upcast 2 dbar averaged data, text files
fsiam06.mat	downcast and upcast data for all casts, matlab file
fsiam0600x.ful	cast x (x=1 to 5) complete data at the full ~2Hz data rate, text files
fsiam0600xful.mat	cast x (x=1 to 5) matlab version of the full ~2Hz data
READMEam06_ctd	a README file detailing file formats

3. Station and summary info files

The files station.dat and borehole_data_summary.dat are updated to include the latest season's data.

4. Current meter files

Matlab files, named dopam0500x.mat and dopam0600x.mat, provided as is (no further information).